ENGINEER'S WALLAH

CHEMISTRY

Matter and Its Classification - I

Lecture No. -1



By: Sunil Bhaiya

TOPICS *to be covered*

1. Introduction to Matter
2. Physical Classification of Matter
3. Interconversion of States of Matter

Class Content Flow and Rules

Concepts to be taught today (Knowledge Ride)

Glossary (English to Hindi)

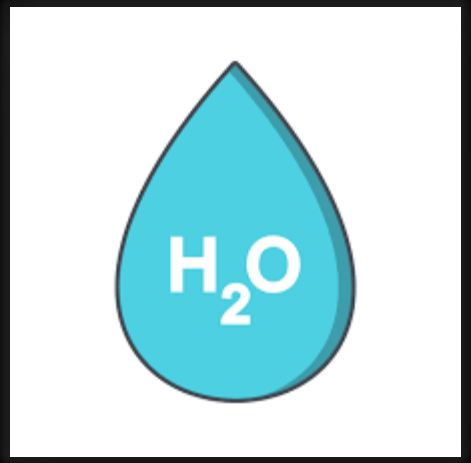
WHAT?

Brief explanation of concepts (Concept Bites)

Practice questions (Give a Thought and Practice and PYQs' Wallah)

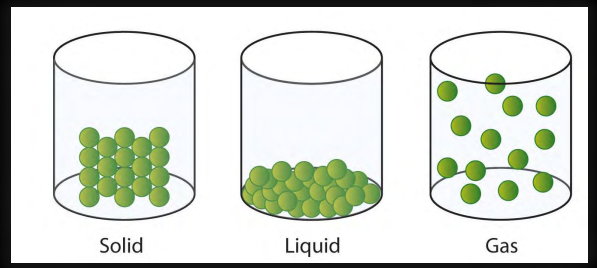
Tips to increase study efficiency (Efficiency Hacks)

Knowledge Ride On



Introduction to Matter

Knowledge Ride On



Physical Classification of Matter

Knowledge Ride On



Interconversion of States of Matter

Knowledge Ride On



Efficiency Hacks

Glossary

Matter (पदार्थ)
Energy (ऊर्जा)
Volume (आयतन)
Compressibility (दबाव)
Rigidity (दृढ़ता)
Hardness (कठोरता)
Fluidity (द्रवता)
Density (घनत्व)
Temperature (तापमान)
Latent heat of melting/fusion (पिघलने/संलयन की गुप्त ऊष्मा)
Latent heat of vapourisation (वाष्पीकरण की गुप्त ऊष्मा)

Concept Bites – Introduction to Matter

What is Matter?

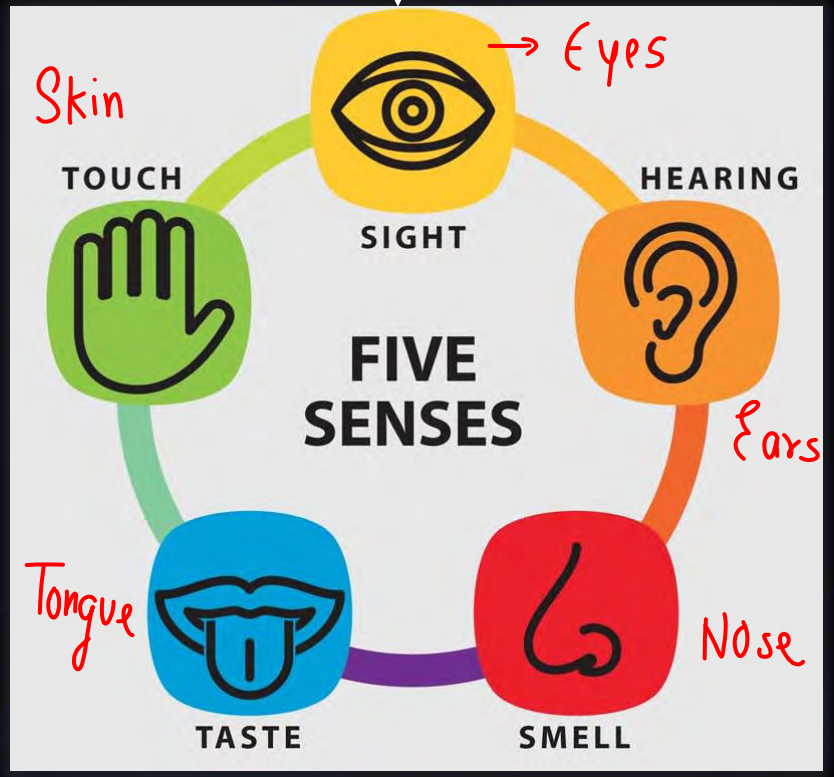
Volume (आयतन)

amount of substance

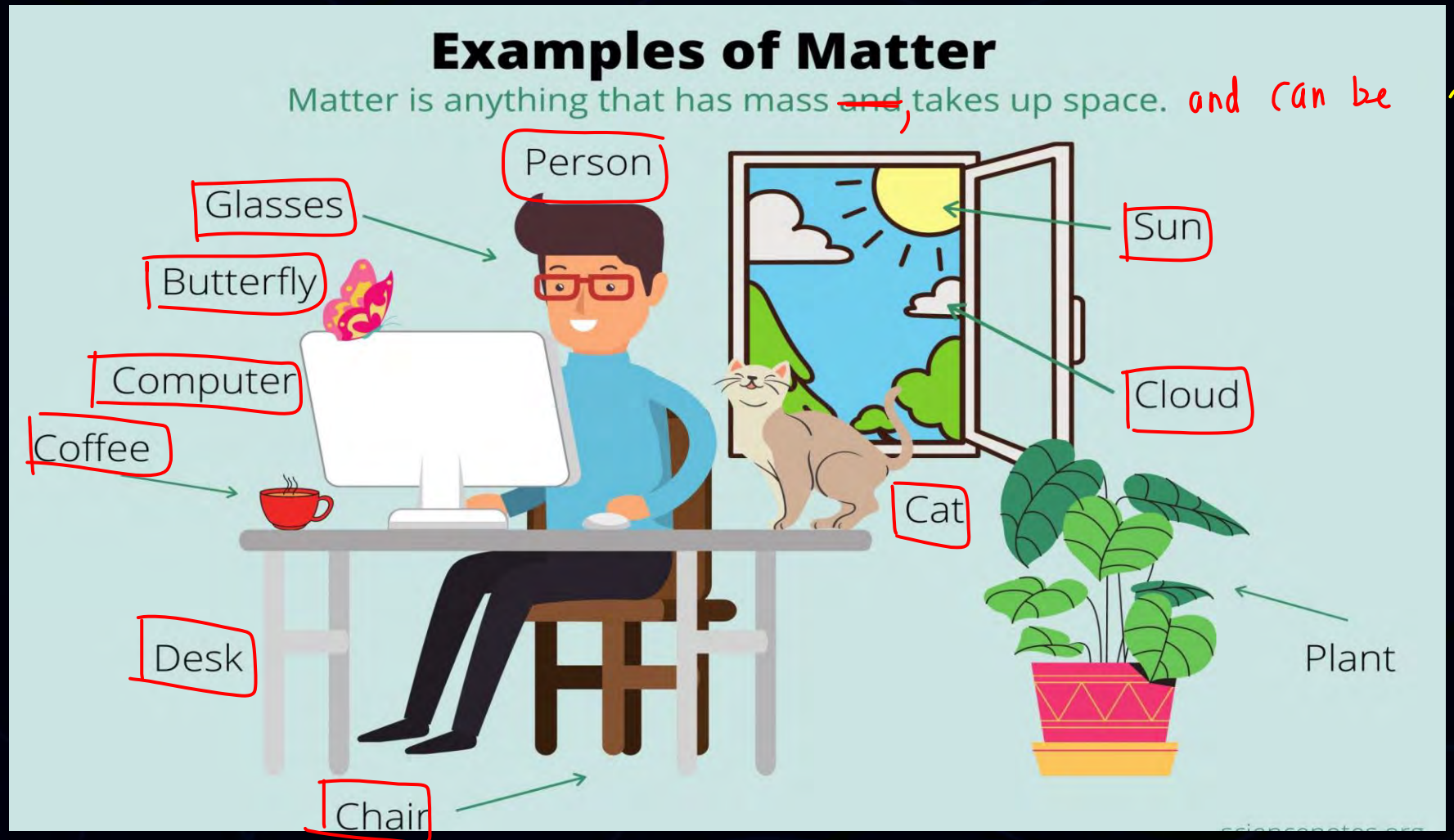
Anything that occupies space, has mass and can be perceived by any one or more of the five basic senses.

(कोई भी चीज़)

महसूस करना



Examples of Matter?



and can be perceived by any 1 or more of 5 basic senses

Give a Thought

Is energy considered to be a matter?!

A Yes

B No

Sound energy, Heat energy, light energy

(HEARING) (TOUCH) (SIGHT)

(i) & (ii) not satisfied

(i) Volume (occupies space)

(ii) has mass

(iii) can be perceived by any one or more of the five basic senses.

Practice and PYQ's Wallah

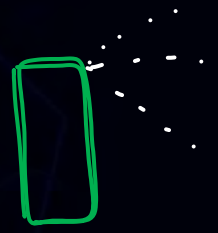


PW Ka ChemStar!

Question

Which among the following can be classified as matter?

		Mass	Volume	Can be perceived by 1 or more of 5 basic senses
A	Thought (विचार)	✗	✗	✗
B	Love (प्यार)	✗	✗	✗
C	Sense of smell (गंध की भावना)	✗	✗	✓
D ✓	Smell of perfume (इत्र की खुशबू)	✓	✓	✓



(प्याज की खुशबू) → Also, matter

Question

Which among the following cannot be classified as matter?

Mass | Volume | Can be perceived by 1 or more of 5 basic senses

A. Nitrogen gas (78.08% in air) ✓ ✓ ✓

B. Almond (बादाम) ✓ ✓ ✓

C. Air (वायु) ✓ ✓ ✓

D. Sound energy (ध्वनि ऊर्जा) ✗ ✗ ✓

Concept Bites – Physical Classification of Matter

Physical Classification of Matter

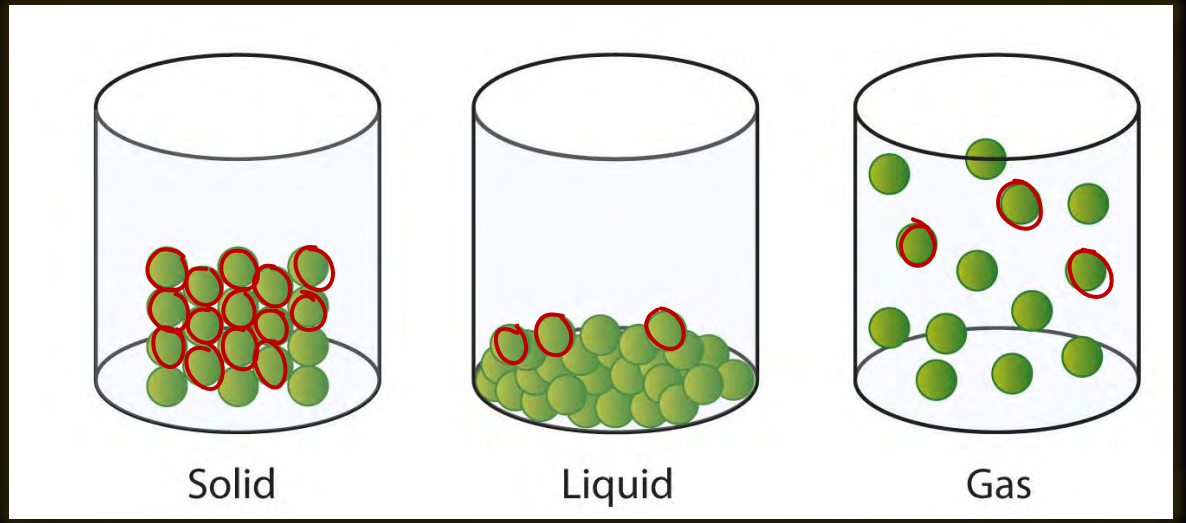
Solid ✓ | Liquid ✓ | Gas ✓

→ Order of force of attraction btw particles of matter

SOLIDS > LIQUIDS > GASES

→ Order of interparticle distance

SOLIDS < LIQUIDS < GASES



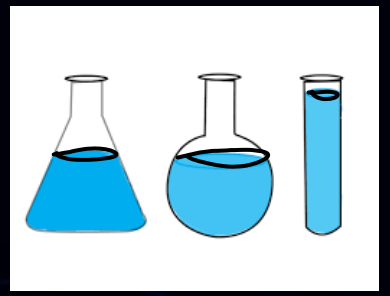
Properties of States of Matter

PROPERTY	SOLID	LIQUID	GAS
Shape and ~~Size~~ Volume (आकार) (आयतन)	Fixed shape and volume	Fixed volume but no fixed shape	No fixed shape and volume

Takes volume of container

CAR







takes shape of container in which it is kept

PROPERTY	SOLID	LIQUID	GAS
Compressibility	Incompressible	Slightly compressible	Compressible

(दबाव)

Compressibility is the measure of how much the volume of a substance especially gases can reduce under application of compressive forces.



PROPERTY	SOLID	LIQUID	GAS
Rigidity / Stiffness	Rigid	Not rigid	Not rigid

'don't bend'

The ability of a substance to maintain its shape without deforming on application of force.

(Form of carbon) Diamond → hardest naturally occurring substance

PROPERTY	SOLID	LIQUID	GAS
Hardness (कठोरता)	Hardest ✓	Not hard ✓	Not hard ✓

The resistance offered by a substance against cutting (काटना) or scratching (घिसना).

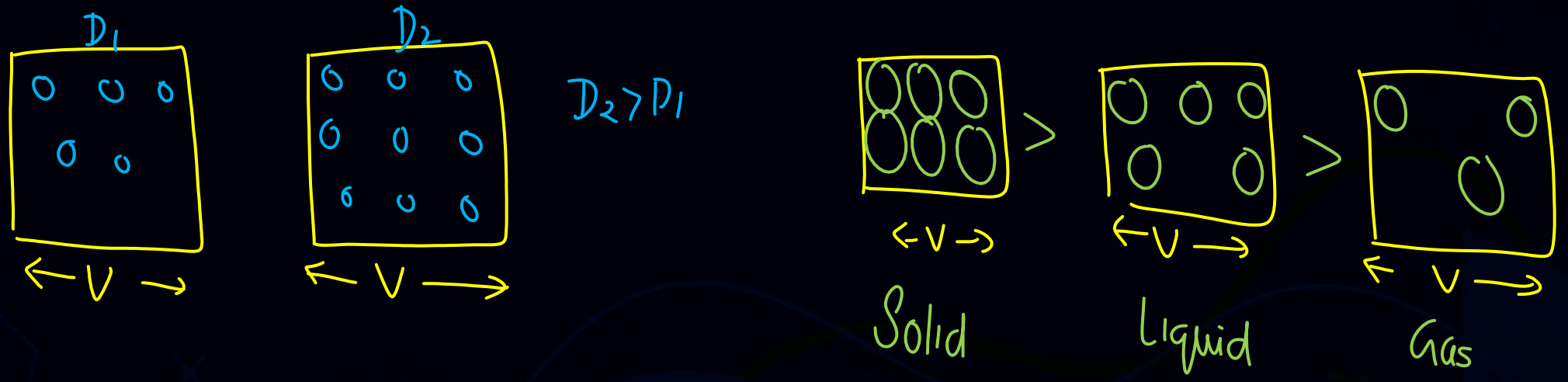
FLUIDS

PROPERTY	SOLID	LIQUID	GAS
Fluidity	Can't flow	Can flow ✓	Can flow ✓

The tendency or ability of any substance to flow. ✓

PROPERTY	SOLID	LIQUID	GAS
Density (घनत्व)	Highest density	Less than solids but more than gases	Less than solids and liquids

The amount of matter (mass) occupied per unit volume.



Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following is the correct order of compressibility of different objects?

A. Chalk (Solid) > Soda water (Liquid) > Nitrogen gas (Gas) (X)

B. Nitrogen gas (Gas) > Soda water (Liquid) > Chalk (Solid)

C. Soda water > Chalk > Nitrogen gas

D. Chalk > Nitrogen gas > Soda water

Question

don't bend or deform

The tendency of solids to maintain their shape under the influence of external force is known as _____.

A Compressibility

B Durability

C Rigidity

D Fluidity

Question

Which among the following is a natural substance that exists in all three physical states?

A. Water

B. Oxygen

C. Hydrogen

D. Table salt

Ice (Solid), Water (liquid), Water-vapour (gas)

Concept Bites – Interconversion of States of Matter

Temperature Scales

- °C
- °F
- K

↳ degree of hotness/coldness of a body

Kelvin (K)

→ SI unit of temperature

$K = 273 + {}^\circ C$

${}^\circ F = \frac{9}{5}{}^\circ C + 32$

(i) 500 K (✓)

(ii) 10 K (✓) → absolute scale of temp

(iii) 0 K (✓) [no negative values (Practically)]

(iv) −30 K (✗)

till now we are unable to achieve it

Practice and PYQ's Wallah



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Question

At what point °C and °F are equal?

A -20

B -40

C +40

D 0

$$^\circ F = \frac{9}{5}{}^\circ C + 32$$

$$\Rightarrow {}^\circ C = \frac{9}{5}{}^\circ C + 32$$

$$\Rightarrow {}^\circ C = \frac{9^\circ C + 32 \times 5}{5}$$

$$\Rightarrow 5\,{}^\circ C = 9\,{}^\circ C + 160$$

$$\Rightarrow -160 = 4\,{}^\circ C$$

$$\Rightarrow \boxed{-40 = {}^\circ C}$$

Let's Practice!

Convert the following from K to °C.

(i) 373 K → 100 °C

(ii) 173 K → -100 °C

$K = 273 + {}^\circ C$

(i) $373 - 273 = {}^\circ C$

$100 = {}^\circ C$

(ii) $173 = 273 + {}^\circ C$

$-100 = {}^\circ C$

Let's Practice!

Convert the following from °C to K.

(i) 273 °C → 546 K

(ii) 25 °C → 298 K

$K = 273 + °C$

(i) $K = 273 + 273$

$K = 546$

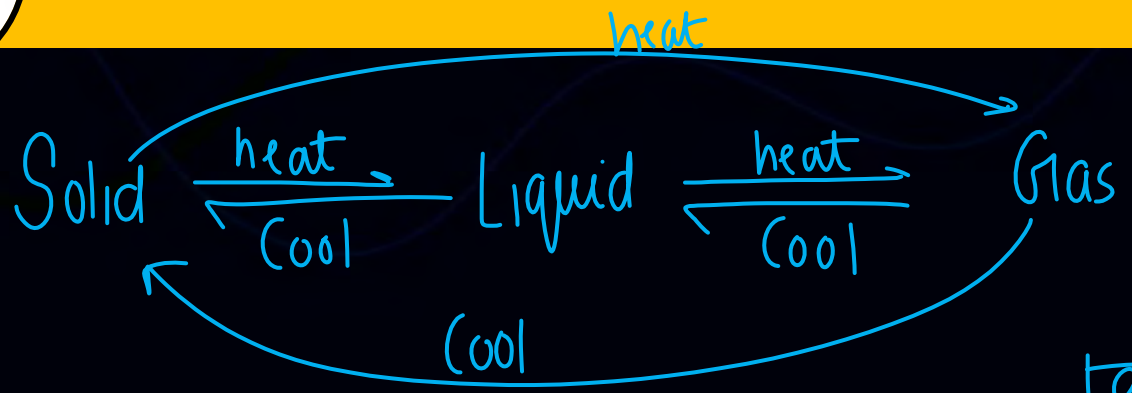
(ii) $K = 273 + 25$

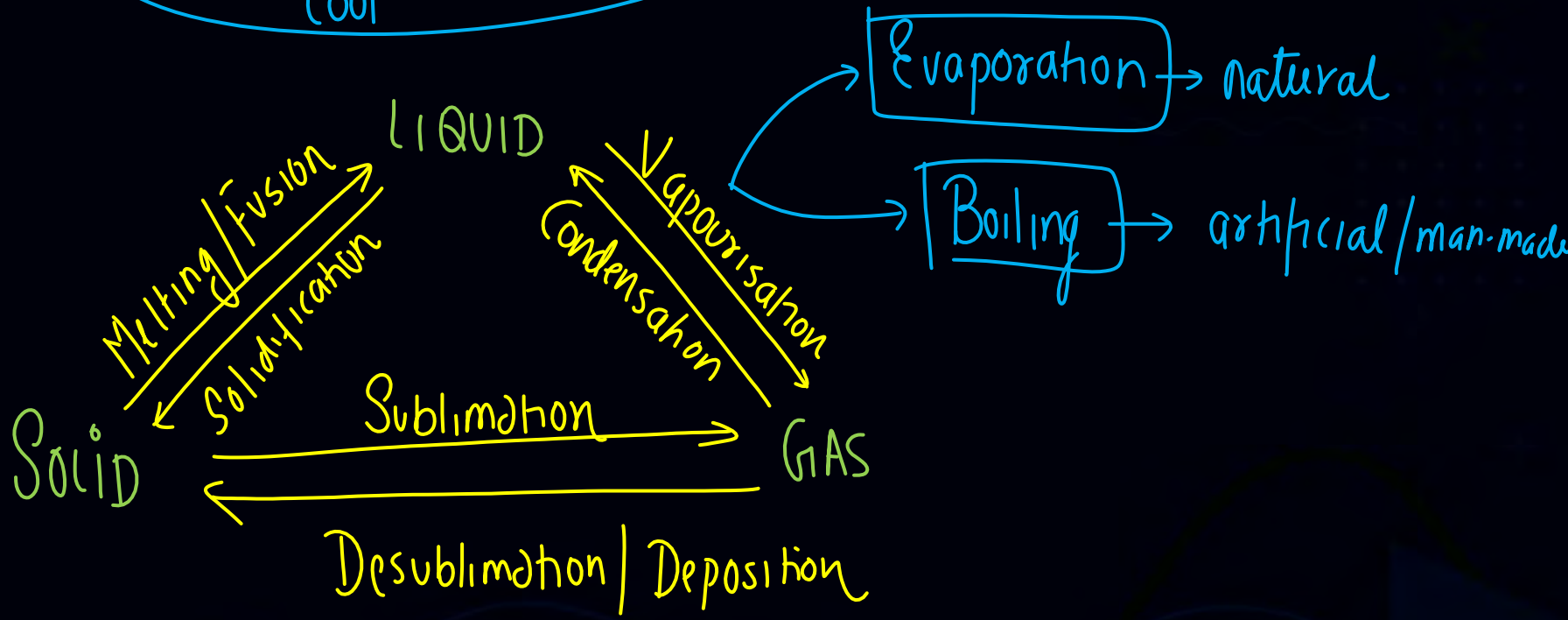
$K = 298$

Room temp



Interconversion of States of Matter





Other Important Facts to Remember

(i) Melting point/fusion point/freezing point of ice is 0 °C.

temp at which ice → water

Ice ⇄ Water (MP / FP)

M.P

F.P → temp at which water → ice

(ii) Latent heat of fusion (melting) of ice: $3.3 \times 10^5 \frac{J}{kg}$

(hidden)

(iii) Boiling point/Condensation point of water is 100 °C.

temp at which water → water-vapour

temp at which water-vapour → water

Other Important Facts to Remember

(iv) Latent heat of vapourisation of water: $225 \times 10^5 \frac{J}{kg}$

Latent heat = supplied at (water → water-vapour), Temperature → constant

Latent heat can't be perceived or measured

(v) Examples of sublimable substances:

Camphor (कपूर), Dry ice (solid carbon dioxide (CO_2)), Anthracene, Napthalene, Ammonium chloride

(vi) A gas can be liquefied by applying high pressure and low temperature. (to make it a liquid)

Other Important Facts to Remember

(vii) Evaporation is a surface phenomenon (particles from the surface of a liquid converts to vapour) while boiling is a bulk phenomenon (particles of entire liquid are involved).

(viii) Evaporation causes cooling while boiling doesn't.

(ix) Evaporation depends on:

(a) Surface area (↑), ∴ Evaporation speed (↑)
(b) Humidity (↑), ∴ Evaporation (↓)
(c) Temperature (↑), ∴ Evaporation (↑)
(d) Wind speed (↑), ∴ Evaporation (↑)

Practice and PYQ's Wallah



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Question

What is dry ice?

RRB JE 2014

A Solid carbon dioxide

B NaCl crystal

C Ice at -50 °C

D Solid H_2O

Efficiency Hack!

Healthy Supper Options

(i) Cowpea Chaat

(ii) Sweet potato and spinach cutlets

(iii) Cornflakes with milk




Thank
You !